

Calibrating the Decision, Not the Model: Per-Class Thresholds for Training-Free Open-Vocabulary Remote-Sensing Segmentation

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Abstract

Training-free open-vocabulary segmentation pipelines built on SAM 3 gate their predictions by a confidence threshold and assign everything below it to a **background** class. We show this discards a great deal of real land cover: 29.68% of all real-class pixels on LoveDA, 323,184,908 in total, with silence outnumbering misclassification 3:1. The size of that residual, and whether it can be detected at all, is governed by the dataset's *label design* rather than by the model. Where **background** is a catch-all covering 36.1% of the scene, the residual is large and no signal separates it from genuine background (nine tested; best AUC 0.622 against a 0.53 floor); where the vocabulary covers the scene (0.84%), only 3.78% is discarded and the runner-up class score separates it at AUC 0.913. Recovery is nonetheless not worthwhile in either regime: it yields +0.04 mIoU on LoveDA, and on OpenEarthMap its apparent +2.28 is entirely attributable to **background** ceasing to be over-predicted (+22.67) while the real classes lose -2.11 IoU. We establish this causally rather than by correlation: because the catch-all's share is set by the vocabulary, we manipulate it directly, and against an arity-matched control detectability falls $0.913 \rightarrow 0.590$ while the control moves to 0.855. A second intervention shows the lever is the label space and not the prompt list. A different correction does work. We fit one threshold *per class* rather than one globally, on 200 labelled tiles, with no model weights trained; this is the protocol the baseline already uses to tune its single threshold. It gives $+1.18 \pm 0.45$ mIoU on LoveDA, capturing 81% of an oracle bound of +1.46, and it is positive on both datasets once the catch-all is excluded from the metric (+1.36 and +1.75 mIoU). **The correction is not specific to SAM 3, but it is specific to the dataset.** Applied unchanged to the nearest published competitor — a CLIP-based pipeline with its own tuned threshold — it adds $+2.51 \pm 0.34$ mIoU on LoveDA with every fold positive, so our method composes with that work rather than competing with it. On OpenEarthMap it helps neither pipeline's headline, and for the same reason in both: the fit must sacrifice a pathologically calibrated catch-all that owns a ninth of the metric. That exclusion is itself a recommendation: mIoU is an unweighted mean, so a catch-all owns $1/N$ of the score however meaningful it is, and we observe it moving the headline in both directions. The correction also demonstrably reaches the residual the first paragraph is about: held out, it lowers the discard rate from 29.25% to 25.01%, and **79.2% of the pixels it recovers land on the correct class — 0.26 wrong per right, against 1.73 for the global threshold sweep that reaches the same mass.** A *second* correction follows from the same argument. Per-class thresholds are provably everything available *after* the argmax, so we also fit one multiplier per class *before* it, on the same tiles and still training nothing. This reaches errors no threshold can: Potsdam's **tree**, which has the largest precision-recall gap we measure (+54.7) and which thresholds move +0.32, gains +21.17 IoU. The two levers turn out to be **substitutes** — the scale adds +4.86 mIoU where thresholds add +0.59, and -0.10 where they add +2.51 — so which one pays depends on the pipeline's operating point rather than on its backbone. On a fourth dataset — low-altitude drone video, a different sensor from the other three — both levers hold under folds grouped by flight, and applied unchanged across a split boundary they raise 56.86 to 63.55, verified in the unmodified evaluation pipeline against a cached prediction that agrees exactly. **We then found that most of the second lever's gain there was repairing a word.** The vocabulary shipped with the baseline names that dataset's *low vegetation* class **vegetation**, which also describes trees; correcting the single word is worth +3.53 mIoU and removes 64% of the lever's gain, so we report that dataset's contribution as +3.22. **The vocabulary is the largest lever we measure and it has no rule.** The same correction applied to Potsdam's equivalent class *costs* -2.71 mIoU, because the prompt shipped there is already better than the annotation guide's own term; on that dataset **tree**'s recall is unchanged to a hundredth of a point by the rename, so its deficit is visual confusion rather than a naming error and the scale reaches something a word cannot. A one-word change is worth +3.53 on one dataset and +4.94 on another, against +2.32 for our complete method on LoveDA. Every training-free method inherits such a list; none report it. Finally, we show the oracle bound is not reachable without labels, and identify why: eleven label-free rules fail, four of which provably estimate per-class precision, because the optimal threshold solves a *coupled* multi-class objective that no per-class statistic can express.

1 Introduction

Open-vocabulary semantic segmentation promises to label an image against a list of class names in plain English, with no training on the target dataset. For remote sensing this is unusually valuable: annotation is expensive, and it must be redone for every new region and sensor. Recent work replaces the multi-model CLIP stacks that dominated this area with SAM 3, whose presence, semantic and instance heads can be combined into a single pipeline that reports strong results across twenty datasets [17].

That pipeline ends in a threshold. Class scores are gated by a presence score, the highest-scoring class is taken per pixel, and any pixel whose winning score falls below a threshold τ is assigned to **background**. The threshold is tuned per dataset using labels, and everything below it is given up on.

We measure what that costs. On LoveDA, **29.68% of all real land-cover pixels — 323,184,908 of them — are assigned to background**, and across every class discard outnumbers misclassification by roughly 3:1. The dominant failure of this pipeline is not that it is wrong; it is that it is *silent*.

The natural response is to recover those pixels; the natural objection is that a lower threshold would do it more cheaply. Neither works (Table 4). Lowering τ from 0.5 to 0.1 recovers two-thirds of the residual and costs -5.54 mIoU, at 1.73 wrong pixels per right one; removing the presence gate costs -11.97. Our original proposal, a region-level recovery method built on a corpus-level co-occurrence prior, adds +0.2 over copying the largest neighbouring region, where a *perfect* matrix adds +0.3.

Running the same pipeline on a second dataset explains why, and the explanation is the paper’s main contribution. On OpenEarthMap, whose **background** class is a rare genuine label rather than a catch-all, only 3.78% is discarded, there are no catastrophic tiles, and the residual is easy to detect — the runner-up class score reaches AUC 0.913 where on LoveDA nine separate signals sit at chance (the supplementary material).

Detectability is a property of the annotation scheme, not of the model. A catch-all class gives the model a plausible answer everywhere, so a strong second-place score carries no information about whether a real class was suppressed. We do not leave this as a correlation across two datasets. The catch-all’s share is set by the vocabulary, so we *intervene* on it (Section 5.1), against a control that makes the same vocabulary change without raising the share; the effect follows the share and not the class count.

That points somewhere more productive than recovery. One global threshold is serving classes with very different precision-recall profiles: on LoveDA **water** is right 89.5% of the time it fires but finds only 54.7% of what is present, while **agricultural** is balanced. Fitting one threshold per class on 200 labelled tiles, with no weights trained, gives $+1.18 \pm 0.45$ mIoU with the catch-all untouched at -0.01 IoU, so the gain is land cover and not a repaired catch-all.

Contributions.

1. **A mechanism for the residual’s size, tested as a forecast.** The catch-all’s share of ground truth orders how much real land cover a presence-gated pipeline discards, monotonically across three datasets (Table 1); we computed the third from ground-truth masks alone and registered it before running the pipeline. We then manipulate that share through the vocabulary, with class count controlled, and the effect follows the share (Table 2); a further intervention localises it to the label space rather than the prompt list. **We also report a registered prediction of ours that failed:** the same variable does *not* order detectability across datasets.
2. **A method.** Calibrated per-class thresholds: $+1.18 \pm 0.45$ mIoU on LoveDA, catch-all -0.01, 81% of an oracle bound of +1.46, verified end-to-end in the pipeline, and positive on *both* datasets on the catch-all-excluded metric (+1.36, +1.75). **It is not backbone-specific but is dataset-specific:** on a CLIP-based competitor at its own operating point it gives $+2.51 \pm 0.34$ on LoveDA and nothing on OpenEarthMap, exactly as for our own pipeline (Table 5). **The rule is also the complete family for this decision:** any per-class monotone recalibration composed with a global threshold reduces to it exactly (Sec. 6.1).
3. **A second lever, and the bound that motivated it.** Because the completeness result above holds only *given* the argmax, we fit a per-class scale *before* it (Sec. 6.2). It adds $+4.86 \pm 0.35$ mIoU on Potsdam, $+1.16 \pm 0.19$ on LoveDA and $+5.89 \pm 1.51$ on UAVid over thresholds alone, all verified end to end, and -0.10 on the CLIP pipeline. **We pre-registered a positive result there and were wrong;** the mechanism we proposed is withdrawn in Sec. 7.2.
4. **A larger lever than the method, and a correction to our own result.** Every training-free method inherits a hand-written prompt list and none report it. Replacing one word with the dataset’s own class name is worth +3.53 mIoU on UAVid — more than our complete method earns on LoveDA (+2.32) — and it removes 64% of the second lever’s gain there, so we restate that dataset’s contribution as +3.22 (Sec. 7.4). **The same rule applied to Potsdam costs -2.71 mIoU**, so the annotation

guide’s term is not automatically the better prompt and no lookup rule exists. Both arms were registered before running; one prediction of each pair failed and is reported.

5. **A reporting recommendation.** We observe the catch-all moving the headline mIoU in both directions, supplying +22.67 IoU of an apparent gain in one experiment and cancelling a real one in another, and recommend reporting mIoU over the real classes beside it.
6. **A bound, with its reason.** eleven label-free rules fail to reach that bound. Four provably estimate per-class precision, which we had supposed was the missing quantity; it is not.
7. **A null-model correction** for co-occurrence analysis on segmentation masks, which refuted one of our own earlier findings and generalises beyond this setting.

Every negative result is reported with an oracle bound, meaning what a perfect version of the missing component would have bought, so a reader learns where the headroom is rather than only that we failed to reach it.

2 Related work

Training-free open-vocabulary segmentation for remote sensing. SegEarth-OV [16] established the setting, introducing a feature upsampler trained on unlabelled imagery to recover the spatial detail that CLIP’s 1/16 downsampling destroys. Its successor, SegEarth-OV3 [17], replaces the CLIP stack with SAM 3 [2] and contributes two components we inherit: *dual-head mask fusion*, which takes the elementwise maximum of the dense semantic head and the aggregated instance head, and *presence-guided filtering*, which multiplies each class map by a scalar presence score before thresholding. It is our baseline throughout, and we reproduce it to within rounding on both datasets.

Dense prediction from vision–language models. The training-free line begins with the observation that CLIP [21] carries usable dense information once the pooling that destroys it is removed [28], and continues through corrections to its attention [22, 13] and to the features guiding it [14]; grouping-based alternatives learn segment structure from text supervision alone [25]. SAM 3 changes the setting rather than extending it: presence, semantic and instance heads arrive in one model, so the pipeline collapses to prompting and thresholding. Its instance head inherits DETR’s [3] set prediction and MaskFormer’s [5] mask classification, its semantic head is FCN-style [18], and the promptable setting is SAM’s [10]. Our results concern what that final threshold does, so they apply to any pipeline in the family that ends in one.

Thresholds and calibration. The step we study is a decision rule over per-class scores, and that such scores are systematically miscalibrated is well established [8]. What is specific here is that the pipeline is training-free, so the usual remedies are available only in the restricted sense the baseline’s own threshold already is: a small labelled calibration set and nothing more. We show what that budget buys and bound what can be done without it. Classical automatic thresholding [20] is among the label-free rules we test, and it does not reach the bound.

Context at inference. ConInfer [4] is the nearest competitor: it fits a Gaussian mixture over self-supervised patch features [19] and fuses the posterior with the vision–language prior, reporting +2.80 mIoU. Its context is *visual*, at *patch* level, fitted *per scene*; our original proposal was semantic context at region level mined corpus-wide, which ConInfer’s own limitations name as future work. We report it as an ablation rather than a contribution because we measured it and it does not earn its place (Section 6). ConInfer’s gain should *not* be read as evidence that appearance solves the detection problem we identify: it performs context modelling, not residual-class detection. **The two are orthogonal, and we demonstrate it:** ConInfer changes the per-class *scores*, our method changes the *decision rule* applied to them, and applying ours on top of theirs improves their pipeline (Section 7.1).

Spatial context as post-processing. Dense CRFs [11], their differentiable successors [27] and global-context encodings of the label distribution [26] are the classical answer to “use spatial context to fix segmentation”, and remain the baseline a reviewer will raise against any neighbour-propagation scheme. Our findings are relevant to that whole family: we show that on a catch-all dataset the limiting factor is not the propagation rule but the inability to identify which pixels should be propagated to at all. The atoms such a scheme propagates over matter more than the rule — we compare connected components against superpixels [1, 7] and find a twenty-point difference in the ceiling (Section 6).

Benchmarks. Both datasets are standard here [23, 24], and a recent benchmark reformulates several under one open-vocabulary protocol [15]. Our reporting recommendation (Section 7) is aimed at such efforts: any benchmark supplying a catch-all hands one class $1/N$ of the headline metric, and we show this is not hypothetical.

3 Setup

We evaluate on LoveDA [23] and OpenEarthMap [24], which differ in exactly the property this paper turns out to depend on: LoveDA’s **background** is a catch-all covering a third of every scene, while OpenEarthMap’s vocabulary very nearly covers it.

Datasets. LoveDA [23] validation, 1669 tiles at 1024^2 and 0.3 m GSD, seven classes including **background**, evaluated at the baseline’s published $\tau = 0.5$. OpenEarthMap [24] validation, 384 of the official 500 tiles, nine classes, at its published $\tau = 0.1$. We reproduce the baseline at 47.38 mIoU on LoveDA against a published 47.4, and 44.19 on OpenEarthMap against a published 42.9. At 77% of the OpenEarthMap split, and with a different tile mix, the latter is a sanity anchor rather than a reproduction gate. A broken class-name to label-value mapping would still have landed far off in one direction, so it establishes that the preparation is sound.

Instrumentation and its validation gate. All measurements come from an instrumented copy of the baseline segmentor that records, per pixel, the winning class and score, the runner-up, the pre-gating fused score, and each head’s own top-1, plus the per-class presence score per sliding-window view. **Every instrumented run must reproduce 47.37 mIoU and 29.68% discard exactly**; a run that does not has changed behaviour and is rejected. This gate is not a formality — it is what establishes that the additions are observation-only, and every number below inherits it.

A cached sufficient statistic. Because the argmax does not depend on τ , the histogram $H[\text{gt, pred, conf bin}]$ is a sufficient statistic for the confusion matrix at *any* per-class threshold vector. Caching it once turns every subsequent threshold sweep, oracle bound and cross-validation into a CPU operation on a saved array rather than a GPU run. We verify it against the deployed pipeline in Section 7.

We run the baseline’s published configuration, and the obvious alternative costs. SAM 3 resizes every input to 1008^2 , so a 1024^2 tile is seen at roughly 1:1 while 512^2 crops would be *upsampled* — the sliding-window inference the rest of this literature uses, and which SegEarth-OV3 does not. We tested it: mIoU falls -3.85, to 43.53. Mean precision *rises* +1.1 while mean recall falls -4.7, and **water** — which occupies part of a tile rather than all of it — loses -14.28 IoU on -16.1 recall.

Because S_{pres} is computed per view, the natural reading is that the gate is at fault: a class absent from one crop is vetoed there, and $P_{\text{final}} = P_{\text{fused}} \cdot S_{\text{pres}}$ then crushes every one of its pixels. **We tested that reading directly and it is wrong.** Gating instead by the maximum presence over crops gives 39.86, and by a single whole-image forward 39.05 — a further -3.67 and worse. Precision collapses as the gate loosens ($68.1 \rightarrow 58.2 \rightarrow 56.7$; building alone $77.0 \rightarrow 49.4$) while recall barely moves. **The per-crop gate was not destroying valid predictions but suppressing false ones**: a class genuinely absent from a crop should be vetoed there, and telling the model it is present somewhere in the tile lets it fire across the whole grid.

The loss is therefore not resolution offset by gating. **Crops cost scene context, that degrades every head, and the per-crop gate partly rescues it** — which identifies a second role for presence gating beyond the suppression of **background** reported in the supplementary: it suppresses classes outside the current field of view, worth -3.67 mIoU here. **The baseline’s whole-image choice is correct, and this is why.** We report the whole sequence because the alternative is standard practice elsewhere, and because the first explanation we reached for was the wrong one.

The multi-view inference that *does* help changes orientation, not field of view. If crops fail because they shrink the scene, the remedy is a transform that does not. Averaging the score stacks over dihedral views of the *whole* image leaves field of view, scene context and presence semantics identical between views, and only the orientation differs — and aerial imagery has no canonical orientation, so a mirrored tile is an ordinary image rather than an unnatural one. A single horizontal flip gives 47.67 mIoU (+0.29); all eight dihedral transforms give 47.84 (+0.47, at 6.59 s per image against 0.85). **Mean precision and mean recall both rise (+0.4, +0.1) — the only intervention we measure for which they do** — and the discard rate does not move, so this keeps the same *quantity* of pixels and a better *selection* of them. Section 7.5 reports what it does and does not compose with.

Table 1: Three datasets, same pipeline, same code. The catch-all’s share of ground truth orders the residual’s *size*. It does **not** order the residual’s *detectability*: Potsdam has a small catch-all and detects like LoveDA, which has a large one. Detection figures are the best cached signal in each case.

	LoveDA	OpenEarthMap	Potsdam
baseline mIoU (published)	47.38 (47.4)	44.19 (42.9)	57.83 (57.8)
background share of GT	36.1%	0.84%	4.29%
real-class pixels discarded	29.68%	3.78%	4.68%
best detection AUC	0.582	0.794	0.578

4 Anatomy of the residual

Two mechanisms, not one. A real-class pixel can be labelled **background** in two ways: its winning score falls below τ , or **background** wins the argmax outright at $\text{conf} \geq \tau$. On LoveDA, 94.0% of the residual is the first and 6.0% the second — 19,378,177 pixels in 24 tiles, *all of them water*, in large homogeneous scenes where the presence head fires for **background**. The distinction matters because **the second group is unreachable by any threshold**: lowering τ cannot change an argmax. Part of the residual lies outside what threshold tuning can address at all.

Threshold relaxation is non-linear, and expensive where it matters. Lowering τ from 0.5 to 0.3 recovers 99M pixels for 0.73 mIoU; from 0.3 to 0.1 recovers a nearly identical count for 4.81. The residual is not a uniform mass: a shallow band sits just below the operating point and a deep tail does not. Tracing the confusion matrix, the full sweep recovers 85.3M correct agricultural pixels while creating 147.9M false ones — 1.73 wrong per right. A scalar threshold carries no information about *which* class a region plausibly is, so it cannot recover selectively.

And the method recovers it selectively, which is the whole claim. Every discard figure above is measured at the published τ , so it is fair to ask how much of the residual the correction of Sec. 6.1 actually returns. Held out over five folds on all 1669 tiles, the discard rate falls from 29.25% to 25.01%: 4.65% of real-class pixels are recovered and 0.40% newly discarded, a net 4.24 points. **Of the recovered pixels, 79.2% land on the correct class — 0.26 wrong per right, against 1.73 for the threshold sweep that reaches the same residual.** The per-class split is the mechanism in miniature: the four classes whose fitted threshold moves *down* recover heavily and accurately (**water** 31.9% $\rightarrow$ 18.6% at 95.0% correct), while **road**, whose threshold moves *up* to 0.675, discards +0.1 points *more* than the baseline and thereby raises its precision. The correction is not a recovery method; it recovers where recovery is cheap and discards harder where it is not.

Presence gating is a correlate, not a cause. A presence-gating hypothesis of our own was refuted by counterfactual and is reported as a correlate only (the supplementary material).

5 The mechanism, and what it does not explain

The residual’s size follows the catch-all’s share. Fig. 1 shows the six quantities that move together. Across three datasets the ordering is monotone: 0.84% of ground truth gives 3.78% discarded, 4.29% gives 4.68%, and 36.1% gives 29.68%. **The third point was predicted before it was measured.** From ground-truth masks alone — no model, no inference — we computed Potsdam’s catch-all share and registered a bracket of 3.78–10.88% with a point estimate of 4.5% in the repository, then ran the pipeline. The measured value is 4.68%.

Detectability does not. The same pre-registration predicted a detection AUC above 0.582, and named the alternative as disqualifying: a dataset near 5% background behaving like LoveDA would mean the mechanism was wrong. That is what happened. Potsdam’s best cached signal reaches 0.578, against 0.582 on LoveDA, whose catch-all is eight times larger. The base rate among catch-all-assigned pixels is 75.1% on Potsdam, between LoveDA’s 43.1% and OpenEarthMap’s 82.7%, so this is not an artefact of class balance.

We therefore split the claim. The catch-all’s share of the label space governs how much real land cover a presence-gated pipeline discards — three datasets, monotone, one of them forecast in advance. It does *not* determine whether that residual can be detected: OpenEarthMap at 0.84% reaches 0.794 while

Label design, not the model, governs the background residual

A catch-all 'background' class gives SAM 3 a plausible answer everywhere: the residual is large and indistinguishable from genuine background. Where the vocabulary covers the scene, it is small and easily detected. Panel ① is the cause; ②–⑥ are consequences, each moving as it predicts. $n = 2$ datasets—the direction is consistent, the functional form is not claimed.

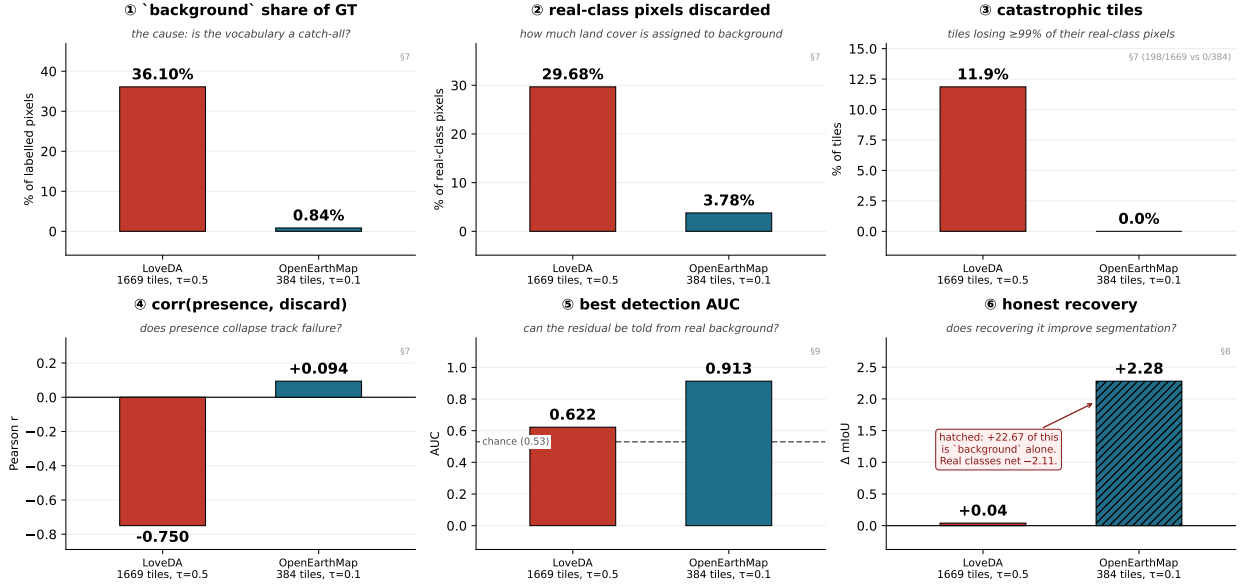


Figure 1: The catch-all class’s share of ground truth as a common cause across our first two datasets. **A third dataset later separated these quantities**: the residual’s size continues to follow the share, and its detectability does not (Table 1). We retain the figure for the two datasets it describes and state the limitation in the text rather than in a caption.

Potsdam at 4.29% reaches 0.578. Something else governs detectability across datasets, and we do not know what it is.

This does not overturn the intervention. Section 5.1 manipulates the catch-all’s share *within* OpenEarthMap and reduces detectability, against a control that changes the class count without changing the share. Both results stand: share is *a* cause of detectability inside a dataset, and is *not* what determines it between datasets. We state that distinction rather than choosing whichever reading is more convenient.

We have deliberately not gone looking for a variable that fits three points. Any account constructed now would be fitted to the observations it must explain, and the value of the confirmed half of this prediction rests entirely on its not having been.

Two of our own claims do not generalise, and we label them. The presence-collapse correlation of Section 4 (-0.750 on LoveDA, 198 catastrophic tiles) becomes +0.094 with 0 catastrophic tiles on OpenEarthMap. It is a property of LoveDA’s annotation scheme, not of SAM 3. Likewise **29.68% is not a general figure** — the identical pipeline discards 3.78% on OpenEarthMap and 4.68% on Potsdam.

Confusability is not the driver. A stratification within LoveDA, where share and confusability dissociate, eliminates confusability as the driver; it is superseded by the intervention above and given in the supplementary material.

5.1 Catch-all share, intervened on

The evidence so far is observational: across datasets and across LoveDA’s urban and rural strata, detectability tracks the catch-all’s share. Stratification cannot rule out everything those strata differ in. But catch-all share is not a property one has to go looking for — **it is set by the vocabulary**. So we manipulate it.

Every class in the baseline is an independent forward pass with its own text prompt, and the only cross-class operation in the pipeline is the **argmax**. Dropping a class from the vocabulary is therefore *exactly* equivalent to dropping its channel, and one cached score stack answers every vocabulary question. All 13 arms read identical model outputs, so they differ in the vocabulary and in nothing else — there is no run-to-run variation to absorb.

Table 2: Raising the catch-all’s share from 0.84% to 58.20% of ground truth, against an arity-matched control that applies the same vocabulary change without raising the share. Detectability is $\max(\text{AUC}, 1 - \text{AUC})$.

signal	published	dose	control	ratio
conf	0.794	0.582	0.710	$2.5\times$
conf2	0.913	0.590	0.855	$5.6\times$

We raise the catch-all’s share by removing classes from the vocabulary and labelling their pixels catch-all, which leaves the catch-all its own single weak prompt. This is the situation a dataset like LoveDA presents. **Each dose arm is paired with an arity-matched control** that removes the same prompts but labels those pixels a real class, so the class count changes identically and only the share differs. Without that control the experiment measures “fewer classes is an easier problem”.

Detectability must be scored direction-agnostically. Detectability is scored as $\max(\text{AUC}, 1 - \text{AUC})$ throughout, because AUC is symmetric and an inverted detector is not an absent one; the supplementary material gives the arm that forced this and the design flaw it exposed.

Both signals degrade with the share, and neither degradation is explained by the class count (Table 2). The intervention moves OpenEarthMap from its own 0.794 into the 0.58–0.62 band where LoveDA sits, reproducing the observational regime rather than merely correlating with it. Against the weakest of the three controls rather than the matched one, the runner-up effect is still $3.1\times$.

The label space is the lever, not the prompt list. A second intervention separates the two things the dose arms change together. Removing the catch-all from LoveDA’s *vocabulary* while leaving its *label space* untouched — 35% of ground truth is still catch-all, those pixels merely become unnameable — moves detectability from 0.592 to 0.540, i.e. not at all. Across both datasets, **every arm that changes the vocabulary without changing the catch-all’s share of ground truth moves at most 0.052, while every arm that raises that share moves 0.582–0.590**. The practical consequence is that a catch-all-heavy benchmark cannot be repaired by editing the prompt list.

The effect saturates, which bounds the claim. The effect saturates by roughly a third of the scene, which bounds the claim and explains why the reverse arm has no room to act; the supplementary material gives the doses.

6 The method

6.1 Lever one: per-class thresholds

A single threshold is being asked to serve classes with very different operating points. On LoveDA, **water** has precision 89.5% against recall 54.7%: it is right nine times in ten when it fires and finds half of what is there, so it can afford to fire far more readily. **agricultural**, at 66.9% against 62.0%, cannot.

We therefore fit one threshold per class by coordinate ascent on a calibration subset of labelled tiles, and evaluate on disjoint tiles (Fig. 2). **No model weights are trained.** The baseline already tunes its single τ per dataset using labels, 0.5 for LoveDA and 0.1 for OpenEarthMap, so this is the same protocol with $N - 1$ free parameters instead of one. The comparison is parity rather than a leak. The published- τ baseline is recomputed on the same held-out tiles.

The objective excludes the catch-all. We fit to maximise mIoU over the real classes only. This matters: optimising *full* mIoU allows the search to buy the metric by repairing an over-predicted background at land cover’s expense, which on OpenEarthMap trades **road** and **building** away for a large **background** gain. Full mIoU is still what we report.

Per-class thresholds are the complete family, not one option within it. It is natural to ask why we do not instead *calibrate* the scores — Platt scaling, isotonic regression, a per-class temperature — and keep the single threshold. The answer is that this would be the same method. Let g_c be any strictly increasing per-class recalibration map. Applying it and then thresholding globally keeps a pixel predicted as class c with score s when $g_c(s) \geq \tau$, which holds exactly when $s \geq g_c^{-1}(\tau)$. Setting $\tau_c = g_c^{-1}(\tau)$ recovers our rule, and setting g_c to any monotone map carrying τ_c to τ recovers theirs. **Every per-class monotone recalibration, composed with a global threshold, collapses to a per-class threshold vector, and conversely.** Fitting a full

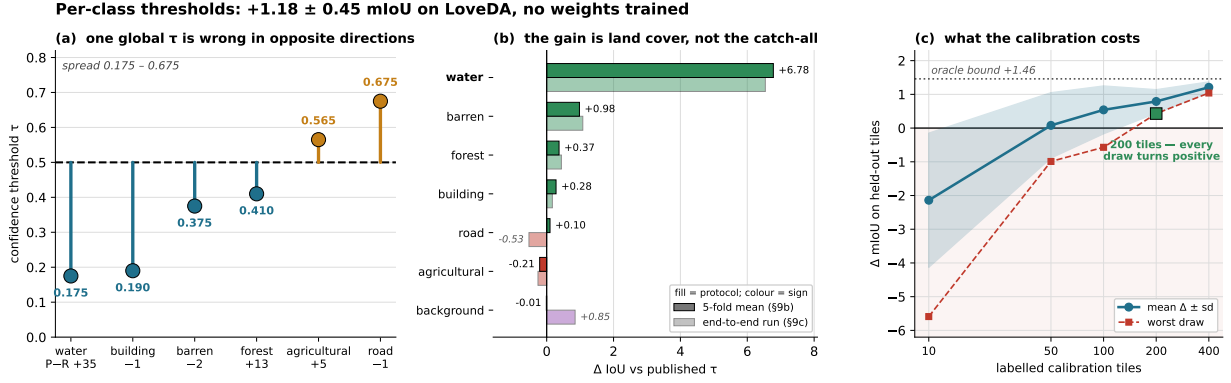


Figure 2: The method. **(a)** The fitted thresholds span 0.175–0.675 against a single 0.5, and move in opposite directions: classes with a large precision–recall gap fire more readily, classes without one fire less. **(b)** The gain is land cover; the catch-all is untouched. Two protocols are shown — five-fold cross-validation and the single end-to-end deployment fit — because they agree on the shape and disagree on **background**. **(c)** The calibration cost: 200 labelled tiles is where every random draw turns positive.

calibration curve therefore spends parameters this decision cannot use: the threshold sees only the single level set $g_c^{-1}(\tau)$, never the shape of g_c . Per-class thresholds are the minimal sufficient parameterisation of the decision, not a heuristic substitute for calibrating it.

And exactly where that argument stops. The equivalence holds *given the argmax*. A recalibration applied *before* the argmax can reorder classes, which is a strictly larger family that no threshold vector can reach: lowering a threshold cannot change which class won. Two error modes live there and neither is addressable by Sec. 6.1’s rule — the argmax-loss mass of Sec. 4 (6.0% of the LoveDA residual, 1.25% of Potsdam’s), and real-class confusion, which is 9.9% of all real-class pixels on LoveDA. The next section closes that family.

6.2 Lever two: scaling before the argmax

We fit one multiplier per class alongside the thresholds:

$$\hat{c} = \arg \max_c w_c \cdot s_c, \quad \text{keep } \hat{c} \text{ if } s_{\hat{c}} \geq \tau_{\hat{c}}. \quad (1)$$

Both vectors are fitted together by coordinate ascent on the same calibration tiles, and no weights are trained. **The threshold reads the raw score $s_{\hat{c}}$, not the scaled one.** That is forced by the completeness argument above: a scale applied to the score the threshold sees is monotone and folds into τ , so reading it raw confines w to the reorderings and makes the gain attributable to them alone.

This is prior correction, and we claim only where it applies. Per-class logit scaling is standard in the long-tail literature. What is untested is whether it helps a training-free open-vocabulary pipeline at an unchanged supervision budget, and — the question that matters here — whether it adds anything *on top of* per-class thresholds. We report it as the increment over Sec. 6.1, never over the published baseline, since the latter would credit the thresholds to the scale.

7 Results

LoveDA. Five-fold cross-validation (Table 3) gives $+1.18 \pm 0.45$ mIoU with every fold positive and a worst fold of +0.80. The gain is **entirely land cover**: excluding the catch-all raises the gain to +1.38 mIoU while the catch-all itself moves -0.01 IoU (+8.30 summed across the six real classes). That makes the result structurally immune to the artefact described below, and it captures 81% of an oracle bound of +1.46 — little is left on the table. It is driven by **water**, which gains +6.78 IoU alone at a fitted threshold of 0.195 against a global 0.5.

The headroom is smaller than it looks, which makes the gain larger than it looks. The baseline paper reports a fully supervised SegFormer-b0, trained on the full LoveDA training set, at 50.0 mIoU. The

Table 3: The method, five-fold cross-validation. Every LoveDA fold is positive. On OpenEarthMap land cover improves substantially while full mIoU stays flat; see the discussion below.

	LoveDA	OpenEarthMap
Δ full mIoU, five-fold	+1.18 $\pm$ 0.45	+0.16 $\pm$ 0.93
worst fold	+0.80	—
<i>both metrics, one partition, identical held-out pixels</i>		
Δ full mIoU	+1.16	+0.30
Δ excl. catch-all	+1.36	+1.75
catch-all Δ IoU	-0.02	-11.30
water alone	+6.78 (at $\tau = 0.195$)	—
oracle bound, full mIoU	+1.46	—
share of the bound captured	81%	—

training-free pipeline starts 47.37 and our fit reaches 48.35, so **44% of the gap between training-free and fully supervised is closed by 200 labelled tiles and no trained weights**.

OpenEarthMap is positive too, once the catch-all is separated. Read as full mIoU the method looks flat there, +0.30. Read as catch-all-excluded mIoU it gains **+1.75** — more than LoveDA does. The headline decomposes exactly: +0.30 is +1.56 contributed by the eight real classes and -1.26 contributed by background alone, whose IoU falls -11.30. **The catch-all cancels a real gain, so here the headline understates the method** — the mirror of Fig. 3, where it overstated it. The per-class movements are ordinary segmentation gains: **building** +4.34, **road** +2.94, **pavement** +2.71, **cropland** +2.50. LoveDA is the clean case, where both metrics agree (+1.16 and +1.36) because the catch-all barely moves at -0.02.

Reporting recommendation. We therefore report both, and suggest the field does. That a single averaged score can hide the behaviour it is meant to summarise is not a new worry [6]; what is new here is a concrete, measured instance of it in open-vocabulary segmentation, moving the headline in both directions. mIoU is an *unweighted* mean over N classes, so **a catch-all owns exactly $1/N$ of the metric however meaningful it is** — 14.3% on LoveDA for a class its annotation guide defines as “everything else”, 11.1% on OpenEarthMap for a class covering 0.84% of the pixels. A ten-point move in that one class is worth more than most methods claim. The distortion is present before any method is applied: OpenEarthMap’s **background** sits at 17.13 IoU against a real-class mean of 47.54, so its published headline is depressed by 3.38 points by a single pathologically calibrated class. We have now measured this artefact moving the headline in both directions — inflating it in Fig. 3, deflating it here — which is why we treat it as a property of the metric rather than of either result.

The recommended metric is not one that flatters us. Proposing a secondary metric under which one’s own method looks better is an obvious hazard, so we note that this one has already cost us a result. Our recovery experiment on OpenEarthMap produced +2.28 mIoU — a headline gain, and the only one that approach ever achieved. Measured over the real classes it is -2.11 IoU, with **building**, the baseline’s strongest class, losing -3.75 (Fig. 3). **The metric we recommend eliminated our own most favourable number before it removed anyone else’s**, and it also declines to credit the per-class threshold fit with a headline gain on OpenEarthMap for either pipeline (Table 5). We adopted it because it changed our conclusions, not because it confirmed them.

A third dataset. On Potsdam (2016 tiles, 57.83 baseline against a published 57.8) the fit gives $+0.59 \pm 0.50$ full mIoU across folds — four of five positive, but a spread that covers zero, so we report it as *marginal* — and +1.04 on the catch-all-excluded metric, with land cover gaining 5.22 IoU in aggregate. **Land cover improves on all three datasets; the headline follows only where the catch-all is competently predicted**, which is the same pattern Section 7.1 finds across pipelines. Potsdam’s **clutter** sits at 16.96 IoU against a real-class mean of 66.05, depressing its published headline by 8.18 points before any method is applied — against 3.38 on OpenEarthMap and 0.31 on LoveDA. One result cuts against us: **car** carries the gain at +3.70 while **tree** gains 0.32 despite a precision–recall gap of +54.7, larger than the **water** asymmetry that drives the entire LoveDA result. The precision–recall statistic did not predict which class would move here.

Table 4: Every intervention we measured. Each is a trivial alternative a reviewer would otherwise propose, and each is reported with an oracle bound where one exists.

intervention	LoveDA	OpenEarthMap
lower τ to 0.1	-5.54	—
remove presence gating	-11.97	—
co-occurrence prior over neighbours	+0.2	—
honest recovery	+0.04	+2.28 [†]
per-class τ , label-free	-0.17	+4.77 [†]
per-class τ, calibrated	+1.18	+0.16

[†] catch-all artefact — see Fig. 3.

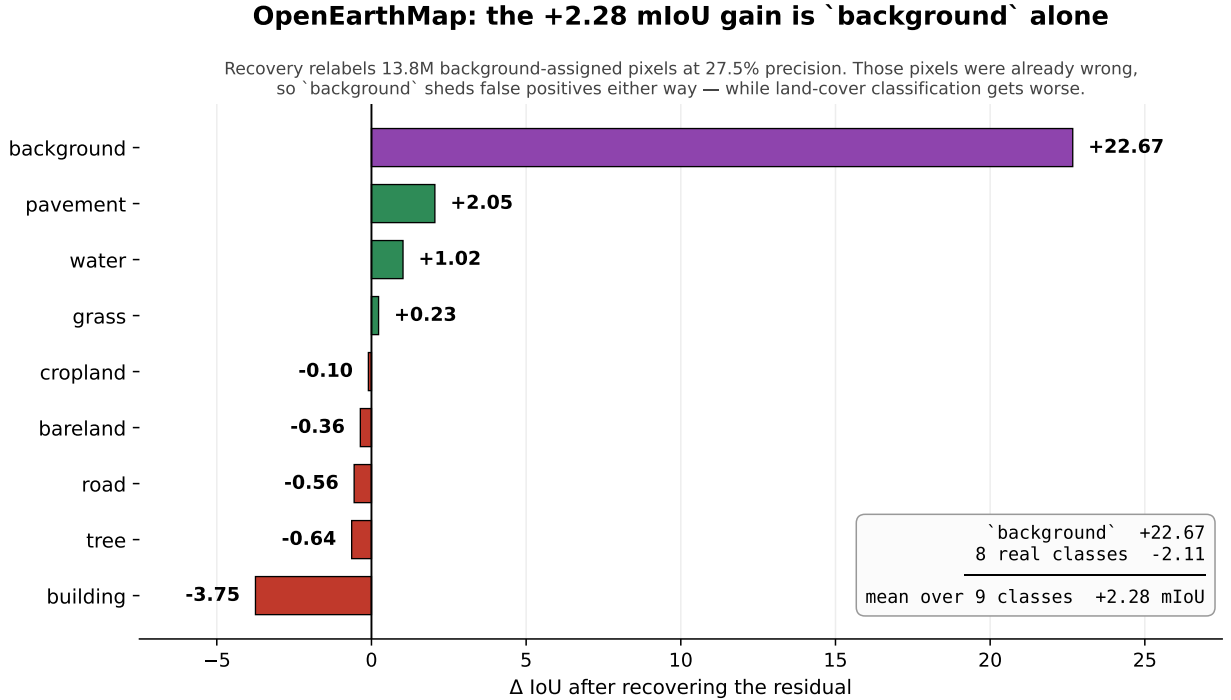


Figure 3: Per-class decomposition of OpenEarthMap’s +2.28 mIoU from recovery. **background** gains +22.67 IoU while the eight real classes lose -2.11 and **building**, the baseline’s strongest class, loses -3.75. A background-assigned real-class pixel is already wrong, so relabelling it either fixes it or leaves it wrong in a different class — while **background** sheds false positives either way. Averaging over nine classes converts a worse segmentation into a better number.

End-to-end verification. The results above are computed from the cached histogram. To confirm that this is a measurement instrument rather than merely arithmetic, we deployed a single fit — 200 calibration tiles — in the actual pipeline. The prediction was 47.16 $\rightarrow$ 48.35 mIoU on 1469 held-out tiles; the pipeline produced 47.16 $\rightarrow$ 48.35, with every per-class delta agreeing to within 0.04. Two honest notes. In this single fit **background** gains +0.85 IoU, 10% of the total rather than the 110% of the artefact below, and incidental — the objective excludes it. And *road loses* 0.53: a per-class rule can hurt a per-class result, which is why we report the full table rather than the mean.

The gain is not uniform, and the thresholds are domain-specific. The gain is not uniform across domains, and the fitted thresholds do not transfer between them; the supplementary material gives the five-fold breakdown and the transfer arms.

7.1 Does it transfer to another pipeline?

A correction fitted on one pipeline invites the obvious objection that it exploits a quirk of that pipeline. We test it on the nearest published competitor, which is CLIP-based rather than SAM 3-based and thresholds its per-class scores at its own tuned values (0.8 on LoveDA). Nothing about the method changes: we cache

Table 5: The same correction, two pipelines and two datasets. Δ full mIoU, five-fold, calibration and evaluation tiles disjoint, each baseline recomputed on its own held-out tiles at its own published threshold. **The axis that separates success from failure is the dataset, not the backbone.**

	LoveDA	OpenEarthMap
SAM 3 pipeline	+1.18 $\pm$ 0.45	+0.30
CLIP pipeline	+2.51 $\pm$ 0.34	-0.39 $\pm$ 1.28
<i>excluding the catch-all class</i>		
SAM 3 pipeline	+1.36	+1.75
CLIP pipeline	+1.94	+1.27

Table 6: The two levers are **substitutes**. Five-fold, all from one script, each dataset at its own published τ . Where the threshold collects a great deal the scale collects nothing, and the reverse.

pipeline	τ	per-class τ	+ scale
ConInfer (CLIP)	0.8	+2.51	-0.10
SAM 3, LoveDA	0.5	+1.16	+1.16
SAM 3, Potsdam	0.1	+0.59	+4.86

its per-pixel scores and run the identical fit.

On LoveDA it transfers, and it is larger. The CLIP pipeline gains $+2.51 \pm 0.34$ mIoU against $+1.18 \pm 0.45$ for our own, with every fold positive, a worst fold of $+2.22$, and *every* class improving ($+2.52$ full mIoU on one partition) — which our SAM 3 result does not manage. It also needs an eighth of the calibration data, 25 tiles against 200. In absolute terms their pipeline moves $37.00 \rightarrow 39.52$ mIoU, so **our contribution composes with the competitor rather than displacing it.**

On OpenEarthMap it transfers to neither, and the failure is shared. The CLIP pipeline gives -0.39 ± 1.28 across folds, one of five positive, and -0.09 on a single partition. One caveat belongs with that row before it is read as a like-for-like test: on this dataset the competitor’s published threshold never fires, because the lowest score their pipeline produces, 0.1042, lies above their published $\tau = 0.1$. Their OpenEarthMap baseline is therefore an unthresholded argmax, and a fit starting from it can only raise thresholds, never lower them — the opposite regime from every other cell in Table 5. But land cover still improves ($+1.27$ excluding the catch-all); what cancels it is **background**, whose IoU falls -10.94 . **Our own pipeline does the same thing on the same dataset**, and to a striking degree: the fit drives that class to 5.83 IoU for SAM 3 and 6.06 for CLIP, from a starting value near 17 in both cases. Two unrelated models, the same sacrifice, the same endpoint.

So the correction is not backbone-specific; it is dataset-specific, and the axis is how well the catch-all is calibrated. Where that class is competently predicted, the fit improves land cover *and* the headline, on both architectures. Where it is pathological — OpenEarthMap’s **background** sits at 17.13 IoU and owns $1/N$ of a nine-class mean — an optimiser told to maximise land cover trades it away, and no headline gain is possible on either pipeline. This is the same conclusion Section 7 reaches from our own results, arriving independently through a competitor’s code.

Two caveats. The deltas are measured on our own reproduction of their pipeline, which reaches 36.99 mIoU on LoveDA against their published 39.33. A delta is robust to a constant offset, but we report the discrepancy rather than only the gain — and the OpenEarthMap reproduction is considerably further off, so we treat that row as supporting rather than independent evidence. On LoveDA their catch-all gains $+6.01$ IoU, about a third of the full-mIoU movement; far from the pathological case of Fig. 3, and the catch-all-excluded column is independently positive, but we quote both as everywhere else.

7.2 The second lever, measured

Two datasets gain, one does not. On LoveDA the scale adds $+1.16 \pm 0.19$ mIoU over per-class thresholds, five folds of five positive, with the catch-all moving -0.03 and the catch-all-excluded figure $+1.36$: both metrics agree, so the gain is land cover. On Potsdam it adds $+4.86 \pm 0.35$, again five of five. On the CLIP pipeline it adds -0.10 ± 0.14 , 1 of 5 folds positive — **nothing**.

End-to-end. Both positives were run through the pipeline, not only the cache. On Potsdam three `eval.py` passes over identical held-out tiles give 57.60, 58.35 and 63.27 mIoU: the scale is +4.92 and both levers together +5.67. **Every rung lands within 0.05 mIoU of its cached prediction and every class within 0.23.** On LoveDA the measured increment is +1.34 against a predicted +1.31.

It does not cost more calibration data. The combined rule fits roughly twice the parameters, so the budget is the obvious objection. It is not larger: on LoveDA 200 labelled tiles remains the point where every random draw turns positive, and on Potsdam 100 suffice. On Potsdam the combined rule at 100 tiles beats thresholds alone at any budget we tested.

tree is the case that shows the mechanism. Potsdam’s `tree` has the largest precision–recall gap we measure anywhere (+54.7: precision 93.34, recall 38.63). It is the clearest instance of the asymmetry Sec. 6.1 exploits, and per-class thresholds move it +0.32 — essentially nothing. The reason is that its residual is below the threshold with a *different* class winning the argmax, so lowering its own threshold returns those pixels wearing the wrong label. At $w = 3.73$ the scale recovers +21.17 IoU, measured by `eval.py`. **A large precision–recall gap says a class is under-firing; it does not say a threshold can do anything about it.**

We predicted the CLIP result wrongly, and report the reasoning that failed. We pre-registered that the scale would add more than +0.50 there, arguing that the gain should track the argmax-loss share. It does not: Potsdam has the *smallest* argmax-loss share of the three (1.25%) and the largest gain. That mechanism is withdrawn rather than patched. What the three datasets show is Table 6’s substitution, and **we state it as a description of three points rather than a law** — it was not pre-registered, and the totals are not constant across them.

7.3 A fourth dataset, and a correction to our own headline

The three datasets above are satellite and airborne imagery. UAVid is low-altitude drone video, a different sensor and a different viewpoint, and its official training split is 200 labelled frames — the same order as the calibration budget the method needs. We reproduce the baseline at 56.86 against a published 54.7, and report both: the +2.16 discrepancy survives identical code, configuration, vocabulary and thresholds, and their released configuration points at a split whose labels are not public, so it is not resolvable from the release.

Two properties of the data change the protocol, and both were initially got wrong. UAVid’s validation split is 7 drone flights of 10 *consecutive* video frames. A frame-level cross-validation therefore places near-duplicates of every evaluation frame in the calibration set, and it reports +1.05 where the group-disjoint answer is +0.51 — **the leak is worth +0.54 mIoU and flips two of five folds**. Its distributed training split further ships 400 flipped and shifted copies among 600 frames, so an image and its own mirror can land on opposite sides of a fold. Both are corrected by grouping folds by flight and excluding the augmented copies; all UAVid numbers here use that protocol. We record them because neither is visible in the directory structure, and both produce a clean, plausible, wrong result.

Both levers work, and the second transfers better than the first. Group-disjoint five-fold over 20 flights gives $+1.34 \pm 0.39$ for thresholds, worst fold +0.75, and $+5.89 \pm 1.51$ for the scale on top — the largest scale gain we measure. Fitting both on the training split and applying them *unchanged* to validation gives +1.03 and +5.64, so the scale retains 96% of its within-split value across a split boundary where thresholds retain 77%. **3 flights are enough:** +0.81 at that budget, with the worst of five draws positive. Shuffling which class receives which multiplier costs -2.82 mIoU and 1.0% of 200 draws match the fitted assignment, so what transfers is the assignment and not the fact that some classes were scaled.

Verified end to end. Deploying both vectors in the unmodified evaluation pipeline gives 57.92 for thresholds and **63.55** for both, against a cached prediction of 63.55 — exact on the headline, every class within 0.20. Pixel accuracy rises $79.24 \rightarrow 84.81$, and mean precision and recall *both* rise (+0.60, +4.92), which distinguishes a better decision from a redistributed one.

Table 7: The same rule, applied to the same class, on two datasets. One word.

	shipped prompt	annotation guide	Δ mIoU
UAvid	vegetation	<i>low vegetation</i>	+3.53
Potsdam	grass	<i>low vegetation</i>	-2.71

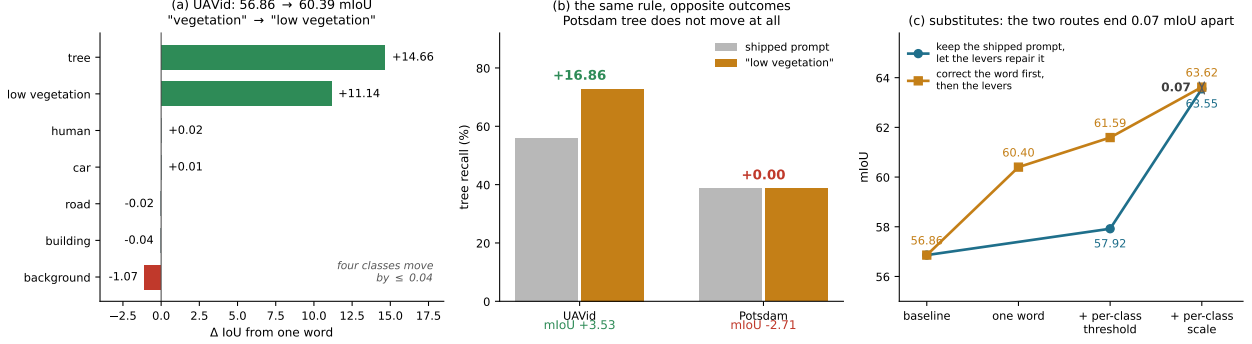


Figure 4: One word against the whole method. (a) On UAvid the correction is surgical: two classes move, four move by at most 0.04. (b) The same rule on Potsdam costs mIoU, and **tree**’s recall does not move at all — the two datasets present identically in a results table and have different causes. (c) The word and the multiplier are substitutes: correcting the prompt first, or leaving it and letting both levers repair it, end 0.07 mIoU apart.

And then we found that most of the second lever’s gain there was a prompt. UAvid’s class is *low vegetation*; the vocabulary shipped with the baseline names it **vegetation**, a word that also describes trees. Correcting that single word (Sec. 7.4) removes 64% of the scale’s gain: $+5.64 \rightarrow +2.03$. **We therefore report UAvid’s method contribution as +3.22, not the 63.55 – 56.86 the uncorrected chain suggests** — $+1.19$ from thresholds and $+2.03$ from the scale, on a corrected baseline of 60.40, reaching 63.62. That is in line with the other datasets ($+2.32$ on LoveDA, $+5.67$ on Potsdam) and it is the figure that survives the check. We report the uncorrected chain too, because the comparison between the two is the finding of the next section.

7.4 A larger lever than either, and no rule for setting it

Every training-free open-vocabulary method inherits a hand-written list of class prompts. None of the work we compare against reports how that list was chosen, and our own results above are stated at a vocabulary we did not write. We therefore tested it, on two datasets, with the prediction committed to version control before each run.

The rule tested is the obvious one, applied uniformly: replace the shipped prompt with the name the dataset’s own annotation guide uses. One word changes; the prompt count per class is unchanged, which matters because prompt *arity* moves scores regardless of meaning (Sec. 5.1).

On UAvid it is worth +3.53 mIoU, and it is surgical. **tree** gains +14.66 and the low-vegetation class +11.14, while *every other real class moves by at most 0.04*. The mechanism is visible in the two numbers we pre-registered: the low-vegetation class was firing on anything green, and its precision rises $53.62 \rightarrow 71.51$; **tree** was losing those pixels, and its recall rises $55.90 \rightarrow 72.76$.

On Potsdam the same rule costs -2.71 mIoU, and we predicted the magnitude wrongly. We pre-registered that the change would move the score by less than one point; it moved -2.71. The shipped **grass** is *better* than the official name — specific, visual, and crucially it does not name the neighbouring class. **So the annotation guide’s own term is not automatically the better prompt, and there is no lookup rule here.**

The prediction that did hold is the one that identifies the mechanism. Potsdam’s **tree** recall is 38.63 before the change and 38.63 after — unchanged to a hundredth of a point — while its precision falls $93.34 \rightarrow 92.17$ as the weaker prompt leaks pixels into it. **Renaming a competing class changes**

Table 8: Five per-class corrections, one protocol and one budget. The two that reshape the *decision* work; the three that reshape the *evidence* the decision reads do not. Gains are over the preceding rung, held out, five-fold.

correction	acts	Δ mIoU
per-class τ	after the argmax	+1.18
per-class scale	at the argmax	+1.16
head fusion	on the score	+0.22
presence weight	on the score	+0.16
per-class bias	at the argmax	-0.07

nothing about tree there. Potsdam’s tree deficit is visual confusion at 5 cm ground sampling distance, not a naming error, which is why the scale reaches it and a better word does not. **The two datasets look identical in the results table — a scale gain carried by tree — and have different causes.**

The word and the multiplier are substitutes. On UAVid the two routes end in the same place: keeping the flawed prompt and letting both levers repair it reaches 63.55, while correcting the word first and then applying both levers reaches 63.62 — 0.07 mIoU apart, and within about one point per class. **Typing two words and fitting a multiplicative reweighting on 200 labelled frames address the same pixels.** One class does not substitute: **human** gains the same amount either way, because it is not losing an argmax to a mis-named neighbour, it is simply too conservative. This is the second substitution we measure between an intervention that adds or corrects evidence and one that fixes the decision.

What we take from this. A one-word change is worth +3.53 on UAVid and +4.94 on LoveDA, against +2.32 for our complete method on LoveDA. **The vocabulary is the largest single lever we have measured in this pipeline, it is unreported, and the direction of a change to it is not predictable from the class name.** We do not offer a method for choosing prompts — our two arms disagree, which is the point. We offer the measurement, and the recommendation that the prompt list be reported and ablated alongside the architecture, because on these benchmarks it is worth more than the methods being compared.

7.5 Three further levers, and what their failure delimits

Per-class thresholds are provably everything available *after* the argmax, and the scale acts *at* it. That leaves the score the argmax reads, and we fitted three further per-class corrections on the same tiles, the same protocol and the same acceptance gate — mean less twice the fold standard deviation above zero, every fold positive — each pre-registered before it was run. Table 8 collects all five.

What the three add. SAM 3 fuses its semantic and instance heads with a hardcoded max, applied identically to every class, although the duality that motivates having two heads is itself class-dependent. We fit that ratio per class. The presence score is a per-class, per-tile ceiling applied with the same uniformity, and classes sit *forty times* apart on it; we fit an exponent per class. Finally, a per-class bias makes the reordering affine rather than multiplicative — vector scaling, the step the calibration literature places immediately after the diagonal case [9, 12] — and it acts precisely where scores are small, which is where the residual lives and where a multiplier cannot reach.

All three are null, and each diagnosed correctly before failing. **The fitted parameters recover real structure; correcting it buys nothing.** The fusion ratio reproduces SAM 3’s things/stuff duality with no supervision about which class is which (Potsdam **car** $\rho = 2.33$ toward the instance head, **building** $\rho = 0.53$ toward the semantic one), and we measure why it cannot pay: for five of six real classes one head wins over 96% of the pixels, so the ratio is a no-op or a uniform rescale, which the scale already fits. The presence exponent orders the classes by how heavily the gate touches them ($\rho = +0.886$, $p = 0.017$, exact), placing **forest** — the most-gated and worst-performing class — lowest. The bias finds one thing consistently, a negative value for the catch-all alone, which reads as *suppress the catch-all where scores are small*; but 6.0% of catch-all assignments are argmax wins and the rest are the threshold rule, so it addresses a twentieth of the mechanism and the scale already reorders that.

What this delimits. Two corrections that change *which label a score earns* succeed; three that change *the score itself* do not. **On this evidence the miscalibration in a presence-gated open-vocabulary**

pipeline is in the decision rule rather than in the per-class evidence, which is a claim about such pipelines in general and not about SAM 3. We state it as a description of five corrections on two datasets, not a law: it is post-hoc, and a post-hoc reading elsewhere in this paper did not survive its first committed test (Section 9). What it does establish is negative and firm — the family of per-class corrections is bounded, and we bound it by testing rather than by asserting.

Adding evidence substitutes for fixing the decision rather than compounding with it. The three corrections above *reallocate* the evidence a pixel already carries. Test-time augmentation instead *adds* evidence — a genuine second measurement — and unlike them it works (Section 3), which makes it the sharpest available test of whether our correction is orthogonal to a better pipeline or compensating for a worse one. Paired fold by fold on identical tiles and identical partitions, augmentation is worth +0.42 mIoU on the baseline, +0.58 once per-class thresholds are applied, and **+0.06 once the per-class scale is**; from the other side, the scale’s own gain falls from +1.26 to +0.74 when augmentation precedes it. **The two are substitutes, and mechanically so: both change which class wins the argmax, while the threshold acts after it and does not compete.** We report this because it bounds our own claim — part of what per-class scaling corrects is score noise that a better measurement would also remove. It removes it far more cheaply, since the scale is fitted once on 200 tiles and costs nothing at inference where augmentation doubles it for the life of the deployment, but a reader is entitled to know the two are not additive.

7.6 The label-free bound, and why it holds

Sweeping a single global τ finds +0.04 mIoU on LoveDA, which is why threshold tuning is usually considered closed. Sweeping a threshold *per class* is worth +1.46 — more than thirty times as much — with the real classes gaining +8.63 IoU. The chosen thresholds span 0.175–0.675: one global value is wrong for different classes in opposite directions.

We tested eleven label-free rules against that bound: three threshold rules (per-class Otsu [20], a confidence percentile, and presence-scaled thresholds) and eight per-class statistics converted into thresholds by a single fitted knob — the same one-parameter budget the baseline spends on its own τ . **None reaches it.** The best scores +0.42 mIoU against a random-proxy control whose 95th percentile is +0.58; on LoveDA every one of the three threshold rules scores *below* the published τ . The control is not optional: one fitted knob over six classes can extract a gain from a meaningless vector.

The reason is not the one we first proposed. We had supposed the oracle exploits per-class *precision*, which is label-derived by definition. That is false. Precision *is* predictable without labels — mean confidence ranks the six classes by precision at $\rho = +0.943$ ($p = 0.017$, exact), and cross-head agreement at $\rho = +0.886$ ($p = 0.033$) — and both move mIoU by less than 0.1. Measuring precision better is not the missing ingredient.

The restated bound. Precision does not determine the threshold. The relation is not even monotone: **water** at 88.5% precision takes $\tau = 0.175$, **road** at 68.5% takes 0.675, and **barren** at 55.6% takes 0.375 ($\rho = -0.429$, $p = 0.419$).

The objective separates, and that is why no scalar locates the threshold. For a *fixed* argmax, a pixel predicted class c either clears τ_c or becomes the catch-all; it can never become another *real* class, because lowering a threshold cannot change which class won. So TP_c , FP_c and FN_c depend on τ_c alone, and the objective is a sum of single-variable terms. We verify this rather than assume it: sweeping each class’s threshold across its full range while the others are held anywhere moves no other class’s optimum by more than 0.000000, over 19 class-sweeps on three datasets. **Coordinate ascent is therefore exact rather than greedy** — one pass per class reaches the joint optimum — and the search costs $N \times 201$ evaluations instead of 201^N .

Separation is what makes the bound bite. Each τ_c is the peak of that class’s own IoU curve, and where that peak sits depends on the whole shape of the class’s score distribution — not on any summary of it. Precision is such a summary, which is why it is measurable and still useless, and why the fitted thresholds are not monotone in it. No per-class scalar locates a peak whose position is a property of a distribution. This explains why an N -parameter fit works and every one-parameter rule fails, and it generalises to any training-free pipeline that thresholds a per-class score.

An earlier version of this argument attributed the bound to coupling between classes — that raising one threshold changes every other class’s optimum. Separation shows that is false for the real classes, and we report the correction because the conclusion survives on a stronger footing. The coupling is real only through

the catch-all: under an objective that scores it, the vector does not separate, and on Potsdam we measure a 0.0050 disagreement there against 0.000000 under the real-class objective.

The same bound governs whether calibration is worth commissioning. The same bound also governs *whether* calibration will pay: stratifying by a label-free residual measure predicts neither the size nor the spread of the gain, on either dataset (the supplementary material). **The calibration cost buys the answer and the question together, and neither is available without labels.**

8 Deploying the correction

Both levers are two lines of arithmetic over scores a pipeline already computes, so what they cost to put into service is worth stating precisely.

What is required. No weights are trained, no architecture is changed, and no second pretrained backbone is downloaded. The complete deployment artefact is two vectors of length N — a threshold and a multiplier per class — fitted by coordinate ascent on 200 labelled tiles in a few minutes of CPU time, and applied at the argmax. On our hardware the fit is negligible beside a single inference pass over the evaluation split. This matters against the alternative a practitioner would otherwise reach for: adapting an open-vocabulary segmenter to a new region by fine-tuning needs labelled data in the thousands of tiles and a training run, and it forfeits the training-free property that made the pipeline attractive. **Our correction asks for roughly a day of annotation and no gradient steps.** It is also additive rather than exclusive — applied on top of ConInfer it raises that pipeline from 36.99 to 39.52 mIoU (Section 7.1), so a deployment need not choose between them.

The vocabulary stays open. Nothing in either lever is tied to a fixed class list: the prompts are read at inference, and the fitted vectors are indexed by position in whatever vocabulary is supplied. A practitioner mapping a class the benchmarks do not contain supplies the prompt and, if labelled tiles exist for it, a fitted entry; if they do not, that class runs at the published global threshold and the remaining classes keep their calibration. **The correction degrades to the baseline class by class rather than all at once**, which is the behaviour a deployment needs when a vocabulary grows faster than its annotations.

What it looks like in practice. Figure 5 shows four Potsdam tiles under both rules, chosen to span the behaviour rather than to flatter it. The first two are gains: a mixed urban scene where **tree** improves while **building** and **car**, already well calibrated, are left within a point of where they were; and a tile where the baseline misses an entire canopy and the correction recovers it. The third is a tile the correction barely touches — fourteen pixels of a quarter-million change label — which is the intended behaviour where nothing is miscalibrated. **The fourth is a tile where the correction loses 7.72 mIoU**: the multiplier that repairs **tree**’s recall across the dataset over-fires on a texturally ambiguous field, and two thirds of the tile changes label with only 26.1% of the changed pixels becoming correct. That failure is intrinsic to the method as specified. The fitted vectors are constant across tiles, so they are right on average — $+4.86 \pm 0.35$ mIoU over five folds, every fold positive — and can be wrong on any individual scene. A per-tile adaptive variant is the obvious next step and we have not tested one.

A vocabulary the benchmarks do not contain. To see what the pipeline does where nothing has been tuned for it, we ran it unchanged on 25 tiles cut from 5 openly licensed UAV scenes over India at 4–5 cm — a region absent from all three benchmarks — under two class lists typed at inference (Figure 6). On a flooded building the two vocabularies recover the same partition to within half a point (78.6%/79.1% for the inundated area under **water** and **flood water**; 18.2%/18.3% for the structure), so **the segmentation is stable under renaming and it is the labels that follow the vocabulary**. On a photovoltaic array the settlement vocabulary, having no word for panels, assigns 100.0% of the tile to a single class; supplying **solar panel** recovers 4.6%.

That last number is our own argument arriving unprompted. The panels are evidently visible to the model — it fires on them — but under one global threshold and no calibration it commits to a small fraction of their extent, which is the low-recall regime per-class thresholds exist to correct (the **tree** case of Section 7.2 at 93.34% precision and 38.63% recall is the same shape). Here we cannot correct it: the imagery is unlabelled, and Section 9 shows thresholds fitted elsewhere would not transfer. We therefore claim nothing about accuracy on these scenes, only that the class list is an input and the model responds to it. The 200 labelled tiles our correction needs are exactly what such a deployment must budget for, and this is what not having them looks like.

Where the correction should not be shipped. The transfer results (Section 9) are a deployment rule, not only a caveat: calibrate on the distribution to be evaluated, and do not pool across domains that differ as LoveDA’s rural and urban halves do. A vector fitted on the wrong domain performs *worse* than not calibrating at all. Because no label-free statistic we tested predicts whether calibration will pay, the honest deployment procedure is to annotate the 200 tiles, fit, and measure on held-out tiles from the same source — which answers both questions at once.

9 Limitations

Two datasets for the mechanism. The method is evaluated on four datasets and a second pipeline; the *mechanism* of Section 5 rests on two. It is established by intervention (Section 5.1) on one of them, and the stratified evidence supporting it comes from the other. A third dataset whose catch-all is common but visually distinct would strengthen both, and we did not have one: the obvious candidate, iSAID, has a catch-all that is 97.11% of ground truth *and* maximally confusable, so it would confirm the ordering and discriminate nothing. The doses in Section 5.1 are chosen greedily from the smallest classes upward rather than at random, and the arity-matched control is not perfectly inert, since dropping a prompt disturbs the scores wherever that class’s pixels lie. We therefore report the dose-to-control *ratio* rather than the difference alone.

Our remaining results are stated at vocabularies we have not audited. Section 7.4 shows a single word is worth +3.53 on one dataset — more than our complete method earns on LoveDA — and that the direction of such a change is not predictable. We tested two prompts on two datasets. LoveDA’s and OpenEarthMap’s lists are untested by the same rule, and we cannot exclude that part of the gains reported for them would also be available by rewording. Two things bound this rather than excuse it. Potsdam was tested precisely because its scale gain has the same signature as UAVid’s and it came back clean, with the shipped prompt beating the annotation guide’s own term. And LoveDA’s list already carries hand-tuned synonyms whose removal costs +4.94 mIoU, which is evidence that it was written with more care than UAVid’s. **Neither is a substitute for auditing it, and we recommend the field report prompt lists as part of a method rather than as part of a configuration file.**

One reproduction does not reproduce. Our UAVid baseline is +2.16 above the published figure under identical code, configuration, vocabulary and thresholds, and we cannot account for it: the released configuration points at a split whose labels are not public, and no preparation procedure is documented. We report both numbers and quote gains against our own reproduction throughout, as we do in the opposite direction for the CLIP-based competitor, whose LoveDA figure we reproduce 2.34 *below* its published value. A gain measured as a difference on one pipeline is robust to a constant offset in that pipeline; an absolute comparison across the two is not, and we do not make one.

UAVid’s validation split is seven scenes. Seventy frames sounds comparable to the other datasets and is not: they are 10 consecutive video frames from each of 7 flights, so the effective sample size is the flight count. Five-fold cross-validation inside it scores on one or two flights per fold and its baseline swings eighteen mIoU points across folds, which is why we fit on the training split and evaluate on validation whole rather than reporting folds within it. The same caution applies to any video-derived benchmark evaluated with frame-level splits.

Calibration must match the evaluation distribution, and the gain is not uniform. Fitting on LoveDA’s *train* split and evaluating on val gives -0.12 mIoU, because those splits differ sharply in discard rate at identical background share. The supplementary material bounds this at a controlled calibration budget: thresholds fitted on one LoveDA domain and applied to the other give -0.40 and -1.11 mIoU, *below* the published τ , so the wrong domain’s calibration is an active harm rather than a diminished benefit. The method requires calibration data from the deployment distribution, and 200 tiles of it; below fifty tiles it actively hurts. Within LoveDA the gain is also concentrated: +2.77 on rural against +0.10 on urban, so our headline +1.18 is a benchmark-level figure and not a claim about every stratum of it. We regard this as the sharpest limitation of the work: the method is a calibration correction, and calibration is exactly the thing that does not transport.

We proposed that the two levers substitute, tested it, and they do not. Table 6 appears to order three pipelines cleanly — where the threshold collects a lot the scale collects little — but three post-hoc

points are not a relationship, and we committed a directional prediction before testing it. LoveDA’s own domains span a $35\times$ range in threshold gain (+2.77 rural against +0.10 urban), so if the levers substituted their scale gains had to differ. They are +1.22 and +1.25: a difference of +0.03 against a standard error of 0.35, where the effect the reading was drawn from spans five points. **This is a refutation from power, not a null from weakness**, and the ordering is withdrawn. What the same measurement shows instead is that the scale gain is *invariant* to what the threshold collected, consistent with the two levers acting on disjoint error mass — thresholds on the 94.0% of the residual that falls below τ , the scale on the 6.0% the argmax loses plus the real-class confusion no threshold reaches. That is two strata of one dataset and inherits the same gate.

The fit captures most of the oracle on two datasets and little on the third. Against the per-class oracle bound, held out, our fitted thresholds capture 83% on LoveDA and 73% on Potsdam but only 11% on OpenEarthMap. The shortfall is one class: OpenEarthMap’s **water** is fitted at 0.710 against an oracle 0.240, costing it $69.18 \rightarrow 55.02$ IoU. Correcting that class alone would take the dataset’s gain from +0.30 to +2.14. It is not a sample-size problem — OpenEarthMap calibrates on a *larger* fraction of its tiles than LoveDA does. We state the figure because it is a measured bound on the method rather than on the benchmark, and a reader would otherwise compute it.

A principled alternative to the fit is worse than the fit. Because the calibration curves are flat near their optima, a one-standard-error rule — take the most conservative threshold within one standard error of the best — is the textbook response, and we named it as the principled choice before running it. It loses: +0.76 on LoveDA against the deployed +1.18, +0.55 on Potsdam and +0.54 on OpenEarthMap, a mean of -0.19 against what we ship. It cannot distinguish a curve that is flat because the class is uninformative from one that is flat near a genuinely displaced optimum, and on LoveDA it drags **water** — the one class that needed moving — back toward the published value. **Selecting which classes to calibrate fails for the same reason, and there we can bound it exactly:** choosing on the evaluation fold, which no deployable rule may do, is worth only +0.08 mIoU, and an honest inner-CV selection gets -0.07. There is nothing to select, so we fit every class.

The scale’s parameters are not uniquely identified. Across folds the fitted multipliers vary by up to 15% on the real classes while the held-out gain holds to ± 0.35 . Coordinate ascent on a coupled objective reaches equivalent optima by different routes, so **the gain is the result and no single fitted vector should be read as “the” scales**. The catch-all’s multiplier varies more still, which is expected: the fit objective excludes it, so it is identified only through the pixels it takes from real classes.

Oracle bounds are bounds under our labeller. Every “oracle” figure we report gives one component perfect information while holding the rest of our implementation fixed. They bound what our design could achieve, not what is achievable in principle.

Scope. OpenEarthMap is evaluated at 384 of 500 tiles. Recovery is evaluated at one threshold per dataset — each dataset’s own tuned value — though Section 7.6 bounds what any other threshold could have given. Deep features remain untested for detection; we judged this unnecessary because detection already succeeds on OpenEarthMap (AUC 0.913) and recovery still fails there, so better detection buys nothing.

10 Conclusion

The residual that presence-gated segmentation pipelines discard is a property of the annotation scheme as much as of the model, and whether it can be detected follows the same axis. Neither regime makes recovery worthwhile: where the residual is hard to find it cannot be found, and where it is easy to find the resulting labels are not accurate enough to beat leaving it alone. What does work is cheaper and less interesting to describe: fitting the threshold per class rather than globally. It works because a single threshold is being asked to serve classes with incompatible operating points. It is positive on both datasets once the catch-all is separated out, and on one of them that separation is the difference between a result and a null. We give the bound on the idea, show it cannot be reached without labels, and identify why: the quantity that sets the right threshold is not a per-class property at all.

The two halves meet in the metric. A catch-all class distorts what is measured as well as what is predicted, and because mIoU is an unweighted mean it does so with leverage $1/N$ regardless of how meaningful that class is. We would rather the field reported the real-class mean beside the headline than continue to discover this one dataset at a time.

We would emphasise one methodological point. Four of the results in this paper are refutations of our own earlier claims, each caught by a test written on purpose — a null model that inverted a published finding of ours, a supervision leak that produced a plausible and quotable number, a detector that scored 0.966 on random noise, and an intervention whose apparent success was a detector changing sign rather than losing information. Reporting a headline mIoU gain without decomposing it per class is, on this evidence, the single easiest way to publish something that is not true.

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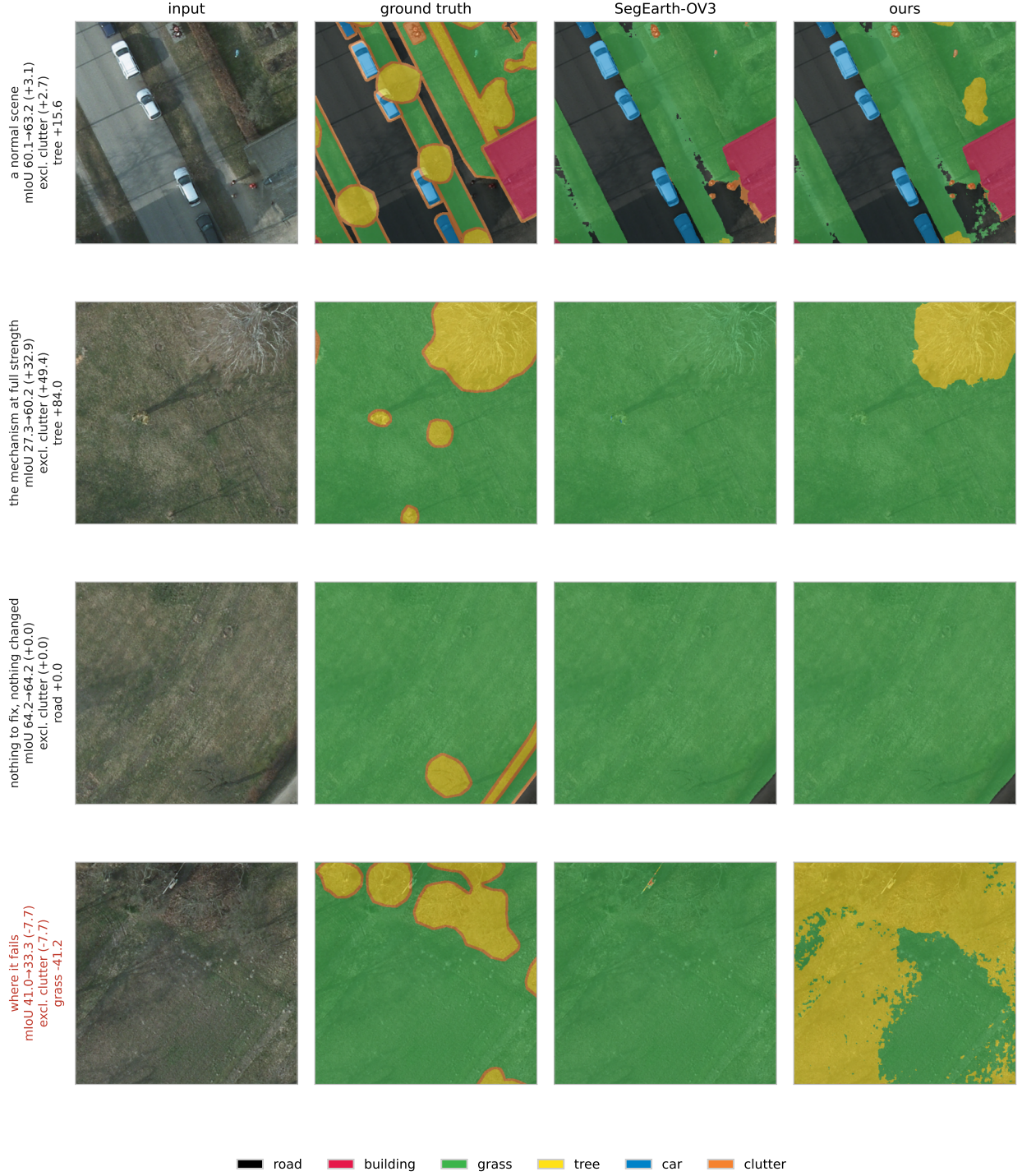


Figure 5: The correction on four Potsdam tiles, chosen to span its behaviour. Same network, same weights, same forward pass: the two right-hand columns differ only in the arithmetic applied to the scores. Rows, top to bottom: a mixed urban scene, where **tree** improves while the already well-calibrated **building** and **car** are left alone; a tile whose canopy the baseline misses entirely; a tile the correction barely touches, fourteen pixels of a quarter-million; and **a tile where it loses 7.72 mIoU** because a dataset-wide multiplier over-fires on an ambiguous field. Per-row: mIoU under both rules, the same excluding the catch-all, and the class that moves most. The fourth row is included deliberately — the five-fold Potsdam gain is $+4.86 \pm 0.35$ mIoU, an average over tiles, and a figure of four gains would misrepresent it.

